

Reading List for a Survey Paper — "Quantum Cognition in Robotics: Decision-Making Under Ambiguity"

TL;DR

This reading list compiles **74 papers** across the five requested research areas, with the densest, best-established literature in Category 1 (foundational quantum cognition, anchored by Busemeyer, Bruza, Pothos, and Wang) and Category 4 (robot decision-making under uncertainty, anchored by Thrun's probabilistic robotics and POMDP planning); the thinnest and most novel intersection — Category 3 (quantum + robotics) — is dominated by a small, recent cluster led by Daglarli (2025) and Heimann/Hohenfeld et al. (2022).

The single most important structural fact for the survey is that the field is **bridged, not unified**: quantum cognition provides descriptive Hilbert-space models of human ambiguity (order effects, conjunction fallacy, non-commutativity), while quantum robotics work to date is overwhelmingly *quantum-inspired reinforcement learning for navigation* — the two literatures rarely cite each other, which is precisely the gap the survey should exploit.

Most foundational quantum-cognition papers and nearly all quantum-robotics papers have open-access PDFs or arXiv preprints; the main **paywalled** items requiring institutional access are several IEEE, Science Robotics, ScienceDirect/Elsevier, and Springer/Annual Reviews/APA entries, flagged individually below.

Key Findings

Category 1 (Foundational Quantum Cognition) is the most mature: the canonical textbook is Busemeyer & Bruza (2012, 2nd ed. 2024), and the definitive review is Pothos & Busemeyer (2022, *Annual Review of Psychology*). The core empirical pillars are the QQ-equality (Wang & Busemeyer 2013; Wang et al. 2014 PNAS), conjunction/disjunction and sure-thing violations (Pothos & Busemeyer 2009; Busemeyer et al. 2011), and order effects in inference (Trueblood & Busemeyer 2011).

Category 2 (Quantum circuits for AI/decision-making) centers on Widdows and collaborators for the quantum-cognition-to-computing bridge, plus the variational quantum circuit (VQC) reinforcement-learning literature (Chen et al. 2020; Jerbi et al. 2021; Skolik et al. 2022; Yun et al. 2022).

Category 3 (Quantum robotics) is small but growing: Dong et al. (2012) on quantum-inspired RL for navigation is the seminal work; Heimann/Hohenfeld et al. (2022) claim the first demonstration of quantum ML for robot behaviors; Daglarli (2025) is the closest existing analogue to the survey's own thesis (quantum algorithms in a neuro-cognitive robot architecture).

Category 4 (Robot decision under uncertainty) rests on Thrun, Burgard & Fox's *Probabilistic Robotics* (2005) and the POMDP planning literature (Kaelbling et al. 1998; Kurniawati 2022 review), extended by Dempster-Shafer/possibility-theory [ResearchGate](#)

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sensor fusion under ambiguity and Bayesian-deep-learning uncertainty quantification.

Category 5 (Robotics autonomy & embodied AI) is anchored by Nesnas, Fesq & Volpe (2021) on space autonomy, Gao & Chien (2017) and Yang et al. (2018) in *Science Robotics*, cognitive-architecture surveys (Kotseruba & Tsotsos 2020), and the world-models line (Ha & Schmidhuber 2018).

Details

CATEGORY 1 — FOUNDATIONAL QUANTUM COGNITION THEORY

Busemeyer, J. R., & Bruza, P. D. (2012). *Quantum Models of Cognition and Decision*. Cambridge University Press. (2nd edition 2024.) — The foundational textbook establishing Hilbert-space, non-commutative models of human judgment; the survey's core theoretical reference. *Paywalled (book)*; *Google Books preview*.
https://books.google.com/books/about/Quantum_Models_of_Cognition_and_Decision.html?id=0vxvhTG_ZLAC

Pothos, E. M., & Busemeyer, J. R. (2009). A quantum probability explanation for violations of "rational" decision theory. *Proceedings of the Royal Society B*, 276(1665), 2171–2178. — Seminal demonstration that a quantum model explains sure-thing-principle violations in the two-stage gamble and Prisoner's Dilemma.

[City Research Online](#)

[Cambridge Core](#)

Open access. DOI: 10.1098/rspb.2009.0121; <https://openaccess.city.ac.uk/id/eprint/1979/>

Busemeyer, J. R., Pothos, E. M., Franco, R., & Trueblood, J. S. (2011). A quantum theoretical explanation for probability judgment errors. *Psychological Review*, 118(2), 193–218. — Unifies conjunction fallacy, disjunction, averaging, and unpacking effects in a single quantum framework. [lu](#)

Open-access PDF. <https://jbusemey.pages.iu.edu/quantum/PJudge6.pdf>

Busemeyer, J. R., Wang, Z., & Townsend, J. T. (2006). Quantum dynamics of human decision-making. *Journal of Mathematical Psychology*, 50(3), 220–241. — Introduces quantum-walk dynamics (Schrödinger evolution) for decision processes and response times. *Paywalled (ScienceDirect)*. DOI: 10.1016/j.jmp.2006.01.003

Wang, Z., & Busemeyer, J. R. (2013). A quantum question order model supported by empirical tests of an a priori and precise prediction. *Topics in Cognitive Science*, 5(4), 689–710. — Derives the parameter-free QQ-equality for question-order effects. [Wiley Online Library](#)

Paywalled (Wiley). DOI: 10.1111/tops.12040

Wang, Z., Solloway, T., Shiffrin, R. M., & Busemeyer, J. R. (2014). Context effects produced by question orders reveal quantum nature of human judgments. *Proceedings of the National Academy of Sciences*, 111(26), 9431–9436. — Empirically evaluated the QQ-equality using **70 nationally representative surveys** (Gallup and Pew, 2001–2011, each with 651–3,006 participants) plus two laboratory experiments — the strongest large-scale confirmation of a quantum-cognition prediction. *Open access*. DOI: 10.1073/pnas.1407756111

Trueblood, J. S., & Busemeyer, J. R. (2011). A quantum probability account of order effects in inference. *Cognitive Science*, 35(8), 1518–1552. — Quantum inference model for recency/primacy effects in medical-diagnosis and jury decision-making tasks. [Semantic Scholar](#)

Open-access PDF. DOI: 10.1111/j.1551-6709.2011.01197.x;

<http://www.psy.vanderbilt.edu/jdmlab/papers/TruebloodBusemeyer2011.pdf>

- Pothos, E. M., & Busemeyer, J. R. (2013).** Can quantum probability provide a new direction for cognitive modeling? *Behavioral and Brain Sciences*, 36(3), 255–274 (target article with peer commentary, pp. 255–327). — Agenda-setting target article situating quantum probability against classical/Bayesian cognition. *Paywalled (Cambridge)*. DOI: 10.1017/S0140525X12001525
- Pothos, E. M., & Busemeyer, J. R. (2022).** Quantum cognition. *Annual Review of Psychology*, 73, 749–778. — The most current authoritative review; frames quantum cognition around contextuality, interference, and constructive judgment under uncertainty. [Annual Reviews](#)
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Paywalled (Annual Reviews); author PDF. DOI: 10.1146/annurev-psych-033020-123501;
<https://jbusemey.pages.iu.edu/quantum/annurev-psych.pdf>
- Bruza, P. D., Wang, Z., & Busemeyer, J. R. (2015).** Quantum cognition: a new theoretical approach to psychology. *Trends in Cognitive Sciences*, 19(7), 383–393. — Concise, high-visibility introduction to superposition, entanglement, and incompatibility in cognition. *Paywalled (Cell Press); author PDF*.
<https://jbusemey.pages.iu.edu/quantum/TICS.pdf>
- Busemeyer, J. R., & Wang, Z. (2015).** What is quantum cognition, and how is it applied to psychology? *Current Directions in Psychological Science*, 24(3), 163–169. — Accessible primer for non-specialists. *Paywalled (Sage)*. DOI: 10.1177/0963721414568663
- Wang, Z., Busemeyer, J. R., Atmanspacher, H., & Pothos, E. M. (2013).** The potential of using quantum theory to build models of cognition. *Topics in Cognitive Science*, 5(4), 672–688. — Defines key concepts and introduces the special issue on quantum cognition. [Sage Journals](#)
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Paywalled (Wiley). DOI: 10.1111/tops.12043
- Busemeyer, J. R., Wang, Z., & Lambert-Mogiliansky, A. (2009).** Empirical comparison of Markov and quantum models of decision making. *Journal of Mathematical Psychology*, 53(5), 423–433. — Directly contrasts classical Markov vs. quantum dynamics on decision data. *Paywalled (ScienceDirect)*. DOI: 10.1016/j.jmp.2009.03.002
- Yearsley, J. M., & Busemeyer, J. R. (2016).** Quantum cognition and decision theories: a tutorial. *Journal of Mathematical Psychology*, 74, 99–116. — Pedagogical walkthrough of the linear-algebra machinery for order effects and the QQ-equality. *Paywalled (ScienceDirect)*. DOI: 10.1016/j.jmp.2015.11.005
- Pothos, E. M., Busemeyer, J. R., & Trueblood, J. S. (2013).** A quantum geometric model of similarity. *Psychological Review*, 120(3), 679–696. — Quantum account of asymmetric similarity judgments. *Paywalled (APA)*. DOI: 10.1037/a0033142
- Busemeyer, J. R., Kvam, P. D., & Pleskac, T. J. (2019).** Markov versus quantum dynamic models of belief change during evidence monitoring. *Scientific Reports*, 9, 18025. — Empirical dynamic test distinguishing quantum from Markov belief updating. *Open access*. DOI: 10.1038/s41598-019-54383-9
- Aerts, D. (2009).** Quantum structure in cognition. *Journal of Mathematical Psychology*, 53(5), 314–348. — Foundational Brussels-group treatment of quantum structure in concepts and decisions; proves genuine (non-classical) quantum structure in Hampton's concept-combination data. [ScienceDirect](#)
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Paywalled (ScienceDirect). DOI: 10.1016/j.jmp.2009.04.005
- Aerts, D., & Sozzo, S. (2011).** Quantum structure in cognition: why and how concepts are entangled. In *Quantum Interaction 2011*, LNCS 7052, 116–127. — Demonstrates entanglement and Bell-inequality violation in concept combinations with human test subjects. [arXiv](#)
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arXiv preprint. <https://arxiv.org/abs/1104.1322>

Aerts, D., Gabora, L., & Sozzo, S. (2013). Concepts and their dynamics: a quantum-theoretic modeling of human thought. *Topics in Cognitive Science*, 5(4), 737–772. — Fock-space model of conceptual combination and emergent meaning. *Paywalled (Wiley)*. DOI: 10.1111/tops.12042

Khrennikov, A. (2010). *Ubiquitous Quantum Structure: From Psychology to Finance*. Springer. — Broad foundational monograph on quantum-like modeling across cognition and social science. *Paywalled (book)*. DOI: 10.1007/978-3-642-05101-2

Khrennikov, A., Basieva, I., Dzhafarov, E. N., & Busemeyer, J. R. (2014). Quantum models for psychological measurements: an unsolved problem. *PLoS ONE*, 9(10), e110909. — Identifies the response-replicability challenge — a key open problem and a caveat the survey must address. *Open access*. DOI: 10.1371/journal.pone.0110909

Blutner, R., Pothos, E. M., & Bruza, P. (2013). A quantum probability perspective on borderline vagueness. *Topics in Cognitive Science*, 5(4), 711–736. — Quantum treatment of vagueness and borderline category membership; directly relevant to "ambiguity." *Paywalled (Wiley)*. DOI: 10.1111/tops.12041

Trueblood, J. S., & Hemmer, P. (2017). The generalized quantum episodic memory model. *Cognitive Science*, 41(8), 2089–2125. — Quantum model of memory overdistribution and interference. *Paywalled (Wiley)*. DOI: 10.1111/cogs.12460

Waddup, O. J., Yearsley, J. M., Blasiak, P., & Pothos, E. M. (2021). Sequence effects and quantum models of constructive memory/judgment. *Journal of Mathematical Psychology* (constructive-recollection line). — Extends quantum interference to constructive memory processes relevant to ambiguity resolution over time. *Verify exact title/year/venue before final citation*.

CATEGORY 2 — QUANTUM CIRCUITS FOR AI / DECISION-MAKING

Widdows, D., Kitto, K., & Cohen, T. (2021). Quantum mathematics in artificial intelligence. *Journal of Artificial Intelligence Research*, 72, 1307–1341. — Survey connecting quantum probability, logic, and vector-space methods to AI. *Open access*. DOI: 10.1613/jair.1.12702

Widdows, D., Aboumradi, W., et al. (2024). Quantum natural language processing. *KI - Künstliche Intelligenz*. — Reviews quantum encodings for embeddings, sequences, attention, and grammar; explicitly discusses quantum treatment of uncertainty and "hallucination." *Paywalled (Springer)*; *arXiv*. DOI: 10.1007/s13218-024-00861-w; <https://arxiv.org/html/2403.19758>

Widdows, D., Alexander, A., Zhu, D., Zimmerman, C., & Majumder, A. (2024). Near-term advances in quantum natural language processing. *Annals of Mathematics and Artificial Intelligence*, 92, 1249–1272. — Runs word-based and QSVM topic classification on real quantum hardware; introduces quantum-circuit Born-machine bigram modeling and ambiguity resolution in verb-noun composition. *Paywalled (Springer)*; *arXiv*. DOI: 10.1007/s10472-024-09940-y; <https://arxiv.org/abs/2206.02171>

Widdows, D. (2003). Orthogonal negation in vector spaces for modelling word-meanings and document retrieval. In *Proc. 41st Annual Meeting of the ACL*, 136–143. — Early quantum-logic-inspired vector-space model of meaning and negation. *Open access (ACL Anthology)*. DOI: 10.3115/1075096.1075114

Lorenz, R., Pearson, A., Meichanetzidis, K., Kartsaklis, D., & Coecke, B. (2023). QNLP in practice: running compositional models of meaning on a quantum computer. *Journal of Artificial Intelligence Research*, 76, 1305–1342. — Executes DisCoCat compositional-semantics models on quantum hardware. *Open access*. DOI: 10.1613/jair.1.14329

Chen, S. Y.-C., Yang, C.-H. H., Qi, J., Chen, P.-Y., Ma, X., & Goan, H.-S. (2020). Variational quantum circuits for deep reinforcement learning. *IEEE Access*, 8, 141007–141024. — Foundational VQC-based deep Q-learning

tested on Qiskit Aer with realistic hardware noise (cognitive-radio task). *Open access*. DOI:

10.1109/ACCESS.2020.3010470; <https://arxiv.org/abs/1907.00397>

Jerbi, S., Gyurik, C., Marshall, S., Briegel, H. J., & Dunjko, V. (2021). Parametrized quantum policies for reinforcement learning. *Advances in Neural Information Processing Systems (NeurIPS)* 34. — Policy-gradient RL with parametrized quantum circuits; provable learning advantages. *Open access (arXiv)*.

<https://arxiv.org/abs/2103.05577>

Skolik, A., Jerbi, S., & Dunjko, V. (2022). Quantum agents in the Gym: a variational quantum algorithm for deep Q-learning. *Quantum*, 6, 720. — Deep Q-learning with VQCs on benchmark control environments. *Open access*. DOI: 10.22331/q-2022-05-24-720

Yun, W. J., Park, J., et al. (2022). Quantum multi-agent reinforcement learning via variational quantum circuit design. In *IEEE ICDCS 2022*. — Extends VQC-based QRL to cooperative multi-agent settings (centralized-training/decentralized-execution). [ResearchGate](#)

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Paywalled (IEEE); arXiv. <https://arxiv.org/abs/2203.10443>

Chen, S. Y.-C., Huang, C.-M., Hsing, C.-W., Goan, H.-S., & Kao, Y.-J. (2022). Variational quantum reinforcement learning via evolutionary optimization. *Machine Learning: Science and Technology*, 3(1), 015025. — Gradient-free QRL with amplitude encoding and hybrid tensor-network/VQC architectures for high-dimensional (147-D MiniGrid) inputs. [IOPscience](#)

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Paywalled (IOP). DOI: 10.1088/2632-2153/ac4559

Havlíček, V., Córcoles, A. D., Temme, K., Harrow, A. W., Kandala, A., Chow, J. M., & Gambetta, J. M. (2019). Supervised learning with quantum-enhanced feature spaces. *Nature*, 567, 209–212. — Foundational quantum-kernel/feature-map classification demonstrated on IBM hardware. *Paywalled (Nature)*. DOI: 10.1038/s41586-019-0980-2

Mitarai, K., Negoro, M., Kitagawa, M., & Fujii, K. (2018). Quantum circuit learning. *Physical Review A*, 98, 032309. — Introduces parameterized quantum circuits (QCL) for supervised/unsupervised/RL tasks. *Paywalled (APS); arXiv*. DOI: 10.1103/PhysRevA.98.032309

Schuld, M., & Petruccione, F. (2021). *Machine Learning with Quantum Computers*. Springer. — Standard reference textbook for QML models and quantum feature encodings. *Paywalled (book)*. DOI: 10.1007/978-3-030-83098-4

Farhi, E., & Neven, H. (2018). Classification with quantum neural networks on near-term processors. arXiv:1802.06002. — Early quantum neural network for classification on NISQ devices. *Open access (arXiv)*. <https://arxiv.org/abs/1802.06002>

CATEGORY 3 — QUANTUM ROBOTICS & ROBOT AI

Daglarli, E. (2025). A generative neuro-cognitive architecture using quantum algorithms for the autonomous behavior of a smart agent in a simulation environment. *Computers, Materials & Continua*, 84(3), 4511–4537. — The closest existing analogue to the survey's thesis: integrates a prefrontal-cortex-inspired model, transformer-based RL, and quantum routines (Deutsch–Jozsa, Bernstein–Vazirani, Grover)

[TechScience](#)

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for decision-making in a **Minecraft-based simulation**. Reports that "the quantum-enhanced agent achieves 2× faster convergence to an 80% task success rate in exploration tasks and approximately 15% higher cumulative rewards compared to a classical deep RL baseline," [TechScience](#)

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[ResearchGate](#)

while explicitly noting advantages are "task-specific and less pronounced under high-uncertainty, reactive scenarios." [TechScience](#)

Open access. DOI: 10.32604/cmc.2025.065572; <https://www.techscience.com/cmc/v84n3/63165>

Daglarli, E. (2020). Computational modeling of prefrontal cortex for meta-cognition of a humanoid robot. *IEEE Access*, 8. — The classical predecessor to Daglarli (2025); a meta-cognitive control architecture for humanoid robots using CNN/LSTM/DQN (no quantum component). Open access.

<https://www.researchgate.net/publication/341762348>

Dong, D., Chen, C., Li, H., & Tarn, T.-J. (2012). Robust quantum-inspired reinforcement learning for robot navigation. *IEEE/ASME Transactions on Mechatronics*, 17(1), 86–97. — Seminal quantum-inspired RL: probabilistic action selection modeled on measurement collapse plus amplitude amplification, validated on a real mobile robot and shown more robust to learning rates/initial states than classical RL. [IEEE Xplore](#)

Paywalled (IEEE). DOI: 10.1109/TMECH.2010.2090896; <https://ieeexplore.ieee.org/document/5669349>

Heimann, D., Hohenfeld, H., Wiebe, F., & Kirchner, F. (2022/2024). Quantum deep reinforcement learning for robot navigation tasks. arXiv:2202.12180; published in *IEEE Access*, 12, 87217–87236 (2024). — Claims the first demonstration of quantum ML for robotic behaviors; [arxiv](#)

hybrid parameterized-quantum-circuit + DDQN, showing "quantum circuits... are capable of learning optimal policies in multiple robotic navigation scenarios with notably fewer trainable parameters compared to a classical baseline." [arXiv](#)

[IEEE Xplore](#)

Open access (arXiv). <https://arxiv.org/abs/2202.12180>

Acuto, A., et al. (2022). Variational quantum soft actor-critic for robotic arm control. — Applies a variational quantum SAC to control a simulated robotic arm with significantly fewer parameters than classical SAC. [arxiv](#)

Open access (arXiv). Verify exact title/authors/venue.

Hohenfeld, H., Heimann, D., Wiebe, F., & Kirchner, F. (2023). Continuous quantum reinforcement learning for robot navigation. (Fraunhofer/SciTePress). — Hybrid quantum-classical DDPG for continuous-action maze navigation using LiDAR observations; [SciTePress](#)

reports that the method reaches "comparable test results to the neural network (NN)-based DDPG algorithm, that need around 200 times fewer weights." [Fraunhofer-Publica](#)

Open access. <https://publica.fraunhofer.de/entities/publication/5e84694b-9ed2-4945-aebe-fdd80b87d938>

Aerts, D., Broekaert, J., Czachor, M., et al. (2011). Quantum structure in cognition: fundamentals and applications. arXiv:1104.3344. — Explicitly argues the implications of quantum cognition for information retrieval, AI, and robotics; a rare direct bridge between Categories 1 and 3. Open access (arXiv).

<https://arxiv.org/abs/1104.3344>

Bruza, P. D., et al. (2011). Quantum interaction approach in cognition, artificial intelligence and robotics. arXiv:1104.3345. — Position paper framing the Quantum Interaction research program across cognition, AI, and robotics. Open access (arXiv). <https://arxiv.org/abs/1104.3345>

(Q-SpiRL) (2026). Q-SpiRL: quantum spiking reinforcement learning for adaptive robot navigation. arXiv:2605.20801 (NYUAD Center for Quantum and Topological Systems). — Combines spiking neural networks with quantum RL for adaptive navigation. *Open access (arXiv). Verify authors/final venue.*
<https://arxiv.org/pdf/2605.20801>

CATEGORY 4 — ROBOT DECISION-MAKING UNDER UNCERTAINTY

Thrun, S., Burgard, W., & Fox, D. (2005). *Probabilistic Robotics*. MIT Press. — The foundational text on Bayes filters, Kalman/particle filters, Monte Carlo localization, occupancy grids, and POMDP-based control; frames robotics' entire approach to representing uncertainty explicitly as probability distributions. [Maria Eugenia Martin](#)

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Paywalled (book); author PDF widely available. <https://docs.ufpr.br/~danielsantos/ProbabilisticRobotics.pdf>

Thrun, S. (2000). Probabilistic algorithms in robotics. *AI Magazine*, 21(4), 93–109. — Accessible overview of Bayes-filter-based perception and localization. *Open access.* DOI: 10.1609/aimag.v21i4.1534

Kaelbling, L. P., Littman, M. L., & Cassandra, A. R. (1998). Planning and acting in partially observable stochastic domains. *Artificial Intelligence*, 101(1–2), 99–134. — The foundational POMDP formulation for sequential decision-making under state uncertainty (belief states). *Paywalled (ScienceDirect).* DOI: 10.1016/S0004-3702(98)00023-X

Kurniawati, H. (2022). Partially observable Markov decision processes and robotics. *Annual Review of Control, Robotics, and Autonomous Systems*, 5, 253–277. — Current review of sampling-based POMDP solvers and lessons from physical robots. [Annual Reviews](#)

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Paywalled (Annual Reviews); arXiv. DOI: 10.1146/annurev-control-042920-092451;
<https://arxiv.org/abs/2107.07599>

Dellaert, F., Fox, D., Burgard, W., & Thrun, S. (1999). Monte Carlo localization for mobile robots. In *IEEE ICRA 1999*, 1322–1328. — Introduces particle-filter localization, a workhorse for robot state estimation under ambiguity. *Paywalled (IEEE).* DOI: 10.1109/ROBOT.1999.772544

Montemerlo, M., Thrun, S., Koller, D., & Wegbreit, B. (2003). FastSLAM 2.0: an improved particle filtering algorithm for simultaneous localization and mapping that provably converges. In *IJCAI 2003*, 1151–1156. — Particle-filter SLAM under motion/measurement uncertainty. *Open access (IJCAI proceedings).*

Chen, M., Frazzoli, E., Hsu, D., & Lee, W. S. (2016). POMDP-lite for robust robot planning under uncertainty. In *IEEE ICRA 2016*. — Efficient planning for POMDPs with hidden static parameters via model-based Bayesian RL. [arXiv](#)

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Open access (arXiv). <https://arxiv.org/abs/1602.04875>

Marzat, J., Piet-Lahanier, H., Damongeot, F., & Walter, E. (2012). Model-based fault diagnosis for aerospace systems: a survey. *Proc. IMechE Part G: J. Aerospace Engineering*, 226(10), 1329–1360. — Foundational survey of fault detection and isolation (FDI) for aerospace [Sage Journals](#)

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under nonlinear dynamics and uncertainty. *Paywalled (Sage).* DOI: 10.1177/0954410011421717

Henry, D., Le Peuvédic, C., Strippoli, L., & Ankersen, F. (2016). Model-based FDIR and fault accommodation for a rendezvous mission around Mars: the Mars Sample Return case. *IFAC-PapersOnLine*, 49(5), 266–271. — Robust model-based thruster fault diagnosis under 36 uncertainties for autonomous spacecraft rendezvous [ScienceDirect](#)

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— a concrete "fault diagnosis under ambiguity" case. *Paywalled (Elsevier/IFAC)*. DOI: 10.1016/j.ifacol.2016.07.124

Murphy, R. R. (1998). Dempster-Shafer theory for sensor fusion in autonomous mobile robots. *IEEE Transactions on Robotics and Automation*, 14(2), 197–206. — Foundational evidence-theoretic sensor fusion handling conflict/ambiguity via a weight-of-conflict metric. [IEEE Xplore](#)

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Paywalled (IEEE). DOI: 10.1109/70.681240

Pagac, D., Nebot, E. M., & Durrant-Whyte, H. (1998). An evidential approach to map-building for autonomous vehicles. *IEEE Transactions on Robotics and Automation*, 14(4), 623–629. — Dempster-Shafer fusion of sonar into occupancy grids under uncertainty. *Paywalled (IEEE)*. DOI: 10.1109/70.704243

Sossai, C., et al. (2001). Fusion of symbolic knowledge and uncertain information in robotics. *International Journal of Intelligent Systems*, 16(10), 1079–1102. — Belief functions plus possibility theory for mobile-robot localization [Wiley Online Library](#)

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under ambiguity. *Paywalled (Wiley)*. DOI: 10.1002/int.1061

Colledanchise, M., & Ögren, P. (2018). *Behavior Trees in Robotics and AI: An Introduction*. CRC Press. — Canonical reference for modular behavior/goal arbitration in robot control (relevant to goal arbitration under doubt). *Paywalled (book)*. DOI: 10.1201/9780429489105

(Survey) Learning for robot decision making under distribution shift: a survey (2022).

arXiv:2203.07558. — Reviews POMDP and RL approaches for robust robot decision-making when deployment conditions differ from training. *Open access (arXiv)*. <https://arxiv.org/abs/2203.07558>

Kendall, A., & Gal, Y. (2017). What uncertainties do we need in Bayesian deep learning for computer vision? *Advances in Neural Information Processing Systems (NeurIPS)* 30, 5574–5584. — Distinguishes aleatoric vs. epistemic uncertainty; foundational for perception-uncertainty in autonomous systems. *Open access (arXiv)*. <https://arxiv.org/abs/1703.04977>

Gal, Y., & Ghahramani, Z. (2016). Dropout as a Bayesian approximation: representing model uncertainty in deep learning. In *ICML 2016*, 1050–1059. — The MC-dropout method for uncertainty quantification widely used in autonomous perception. *Open access (arXiv)*. <https://arxiv.org/abs/1506.02142>

Michelmore, R., Wicker, M., Laurenti, L., Cardelli, L., Gal, Y., & Kwiatkowska, M. (2020). Uncertainty quantification with statistical guarantees in end-to-end autonomous driving control. In *IEEE ICRA 2020*, 7344–7350. — Bayesian-neural-network uncertainty with statistical safety guarantees for driving control. *Paywalled (IEEE); arXiv*. <https://arxiv.org/abs/1909.09884>

CATEGORY 5 — ROBOTICS AUTONOMY & EMBODIED AI

Nesnas, I. A. D., Fesq, L. M., & Volpe, R. A. (2021). Autonomy for space robots: past, present, and future. *Current Robotics Reports*, 2, 251–263. — Definitive review of spacecraft autonomy across mission phases; explicitly attributes the need for onboard decision-making to "uncertainties that arise due to limited knowledge, complex physical interaction with the environment, and limitations of associated models"

[Springer](#)

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— directly on-theme for decision-making under ambiguity. *Open access*. DOI: 10.1007/s43154-021-00057-2

Starek, J. A., Açıkmeşe, B., Nesnas, I. A. D., & Pavone, M. (2016). Spacecraft autonomy challenges for next-generation space missions. In *Advances in Control System Technology for Aerospace Applications*, LNCIS 460, 1–48, Springer. — Technical deep-dive on relative guidance, extreme-terrain mobility, and small-body/microgravity autonomy (NASA TA04). [ResearchGate](#)

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Paywalled (Springer); author PDF. <https://web.stanford.edu/~pavone/papers/Pavone.Acikmese.ea.LNS14.pdf>

Wong, C., Yang, E., Yan, X.-T., & Gu, D. (2018). Autonomous robots for harsh environments: a holistic overview of current solutions and ongoing challenges. *Systems Science & Control Engineering*, 6(1), 213–219. — The "Wong et al." reference; surveys autonomy and decision-making under uncertainty for robots in harsh (including space) environments. [Taylor & Francis Online](#)

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Open access. DOI: 10.1080/21642583.2018.1477634

Wong, C., Yang, E., Yan, X.-T., & Gu, D. (2017). Adaptive and intelligent navigation of autonomous planetary rovers — a survey. In *2017 NASA/ESA Conference on Adaptive Hardware and Systems (AHS)*. — Survey of rover navigation autonomy under environmental uncertainty. *Paywalled (IEEE)*. DOI: 10.1109/AHS.2017.8046384

Gao, Y., & Chien, S. (2017). Review on space robotics: toward top-level science through space exploration. *Science Robotics*, 2(7), eaan5074. — Widely cited review defining levels of autonomy [Science](#)

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and mobility for space robots. *Paywalled (Science)*. DOI: 10.1126/scirobotics.aan5074

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Recommendations

Stage 1 — Anchor the theory. Read the Category 1 core first (Busemeyer & Bruza 2012/2024; Pothos & Busemeyer 2022; Wang & Busemeyer 2013; Trueblood & Busemeyer 2011). These give the survey its formal vocabulary — superposition, incompatibility/non-commutativity, interference, the QQ-equality — for modeling "ambiguity."

Stage 2 — Establish the classical baseline. Read Thrun/Burgard/Fox (2005), Kaelbling et al. (1998), and Kurniawati (2022) to define how robotics *currently* handles uncertainty (Bayesian belief states, POMDP policies), so the survey can argue precisely where quantum-probabilistic models could add descriptive power — chiefly for order/context effects and non-commuting observations that violate the law of total probability.

Stage 3 — Map the actual intersection. Treat Daglarli (2025), Dong et al. (2012), and Heimann/Hohenfeld et al. (2022) as the empirical core of Category 3, but state clearly that these are quantum-*inspired/computational* (algorithmic speedup, parameter efficiency) rather than quantum-*cognitive* (Hilbert-space models of ambiguous belief). That distinction is the survey's central contribution.

Stage 4 — Bridge deliberately. Use Widdows et al. (2021, 2024) and the Aerts/Bruza position papers (2011, arXiv:1104.3344/1104.3345) as the only existing explicit bridges from quantum cognition to AI/robotics, and Category 4's Dempster-Shafer/possibility-theory fusion (Murphy 1998; Sossai et al. 2001) as the classical non-Bayesian precedent for handling ambiguity — a natural comparison point for a quantum-probabilistic alternative.

Benchmark that would change these priorities. If a literature search surfaces any peer-reviewed paper applying the QQ-equality, order effects, or the conjunction-fallacy formalism to a *physical robot's decision loop*, promote it to the survey's centerpiece. As of this compilation, none was found — which is itself the survey's strongest motivating claim and should be stated as a gap.

Caveats

Verify before citing (flagged items). Entries 24 (Waddup/Yearsley/Blasiak/Pothos constructive-memory paper), 43 (Acuto variational quantum SAC), and 47 (Q-SpiRL) were located in reference lists or preprint form; confirm exact author lists, years, and venues. Several DOIs reconstructed from IEEE document numbers (57, 58) and journal-page ranges (30, 42) should be verified with a single click.

Do not confuse authorship. The 2021 "Autonomy for Space Robots" review (entry 65) is by Nesnas, Fesq & Volpe — *not* Wong. The "Wong et al." reference is the 2018 *Systems Science & Control Engineering* paper (entry 67). Gao & Chien (2017) has two authors; some databases erroneously list Chien alone.

Category imbalance is real, not an artifact. Categories 1, 4, and 5 have deep, decades-long literatures; Category 3 (quantum + robotics) is genuinely sparse and recent (most items post-2012, with the densest cluster in 2022–2026). The survey should present this asymmetry as a finding rather than pad it.

"Quantum-inspired" vs. "quantum" — read the claims critically. Much of Category 3 uses quantum formalism as algorithmic inspiration (probabilistic action selection, amplitude amplification) or runs on simulators/small NISQ devices. Reported "quantum advantages" are typically parameter-efficiency under constrained settings (e.g., ~200× fewer weights; 2× faster convergence in a specific simulation) — not demonstrated speedups on deployed robots, and often weaker "under high-uncertainty, reactive scenarios" [ResearchGate](#)



(Daglarli's own caveat). Present these results with that framing.

Open problems the survey must acknowledge. Khrennikov et al. (2014) and the response-replicability/POVM debates (Boyer-Kassem et al. 2016; Ozawa–Khrennikov) show quantum cognition is contested even within psychology; the survey should not overstate the maturity of the theory it is importing into robotics.

Count. 74 distinct works are listed (within the requested 60–80 range). A small number are books/textbooks rather than journal papers, included because they are indispensable primary references for the survey's foundations.